INCH-POUND
A-A-59822
12 February 2009

COMMERCIAL ITEM DESCRIPTION

PLUG ASSEMBLIES, GRIPPERS, OD AND ID TYPES FOR TEMPORARY PRESSURE TESTING OF WATER, OIL AND GAS TRANSMISSION PIPING

The General Services Administration has authorized the use of this commercial item description, for all federal agencies.

1. SCOPE. This commercial item description describes high pressure hydrostatic test plugs and fittings (hereafter referred to jointly as "plugs") that can be used to make mechanical attachments to pipes, tubes, and other piping fittings (hereafter referred to jointly as "pipes"), for the purposes of isolating components and making pressure seal boundaries for hydrostatic testing. As a secondary function, the test plug provides a means for making a connection of the hydrostatic test medium and pressure.

2. CLASSIFICATION. The hydrostatic test plugs shall be of the following types:

Type I – Hydrostatic test plugs that are installed in and seal on the pipe internal diameter.

Type II – Hydrostatic test plugs that are installed on and seal on the pipe outside diameter.

3. SALIENT CHARACTERISTICS.

3.1 Size.

3.1.1 Standard sizes.

3.1.1.1 Type I. Type I hydrostatic test plugs shall be provided in sizes that can be used in the series of pipe diameters and schedules or classes shown in table I.

TABLE I. Pipe sizes for type I hydrostatic test plugs.

Pipe Material	Schedule or Class	National Pipe Sizes – inches
Standard Steel Pipe	40	1, 1¼, 1½, 2, 2½, 3, 3½, 4, 5, 6, 8, 10
	80	1, 1¼, 1½, 2, 2½, 3, 3½, 4, 5, 6, 8, 10
	120	4, 5, 6, 8, 10
	160	1, 1¼, 1½, 2, 2½, 3, 4, 5, 6, 8, 10
	XS	10
	XXS	1, 1¼, 1½, 2, 2½, 3, 3½, 4, 5, 6, 8
Stainless Steel Pipe	5 S	1, 1¼, 1½, 2, 2½, 3, 3½, 4, 5, 6, 8
	10 S	1, 1¼, 1½, 2, 2½, 3, 3½, 4, 5, 6, 8
90-10 Copper-Nickel Pipe/Tubing	200	As defined in MIL-T-16420
70-30 Copper-Nickel Pipe/Tubing	200	As defined in MIL-T-16420
	700	
	1650	
	3300	
	6000	

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FSC 4730

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3.1.1.2 Type II. Type II hydrostatic test plugs shall be sized to apply to the following standard nominal pipe sizes: ¼-inch, ⅜-inch, ½-inch, ¾-inch, 1-inch, 1¼-inch, 1½-inch, 2-inch, 2½-inch, and 3-inch, or as defined by MIL-T-16420.

3.1.1.3 Tolerances. For Type I or II plugs intended for use in standard pipe sizes, the dimensions shall be such that the plug can be properly set and used in pipes with diametrical and wall thickness dimensional tolerance ranges as specified in table II.

TABLE II. Tolerances for standard pipe.

Type of Pipe	Standard for Outside Diameter Tolerance	Standard for Wall Thickness Tolerance
Standard Steel Pipe	ASTM A530/A530M	ASME B36.10
Stainless Steel Pipe	ASTM A530/A530M	ASME B36.19
Copper-Nickel Pipe/Tubing	MIL-T-16420	MIL-T-16420

3.1.2 Non-standard sizes. For plugs required for service in non-standard dimensioned pipes, the dimensional design shall be suitable for use in outside and inside diameter tolerance ranges, and/or wall thickness tolerance ranges, as provided by the purchaser.

3.2 Marking. Hydrostatic test plug assemblies shall be marked by stamping or other permanent means that will not be subject to degradation by corrosion or contact with water. Marking shall be done on a prominent location that does not interfere with the sealing or gripping function. Characters shall be at least ⅛-inch high. All assemblies shall be marked with the following:

a. Manufacturer's name, abbreviation, acronym, or symbol
b. Manufacturer's part number
c. Maximum pressure rating of the plug in pounds per square inch gauge (see 3.4.4)
d. Notation if a safety gag is required (see 3.4.9)
e. Drawing Number
f. Material Type
g. Material Specification
h. Additionally, the following items shall be included in the marking for the specific hydrostatic test plug types:

Type I: The internal pipe diameter range, in inches, over which the hydrostatic test plug can be installed and set properly.

Type II: The Nominal Pipe Size (NPS) diameter, in inches, for which the hydrostatic test plug is intended.

3.3 Materials. All pressure containing materials shall be metallic except for the expandable mechanical stopper material, and any other seals or o-rings required. Materials shall be selected based on a service temperature range of 60 to 120 °F and shall be resistant to corrosion or degradation in fresh water service.

3.4 Design details.

3.4.1 General design. In general, hydrostatic test plugs shall contain: expandable mechanical stopper seals or other sealing devices to seal against the surface of the pipe being tested, gripper assemblies to grip the surface of the pipe in order to ensure against ejection of the plug during hydrostatic testing, and wedging devices activated by tightening the plug fitting(s) in order to activate the gripping elements and the seals. See figures 1 and 2 for a typical OD and ID type design.

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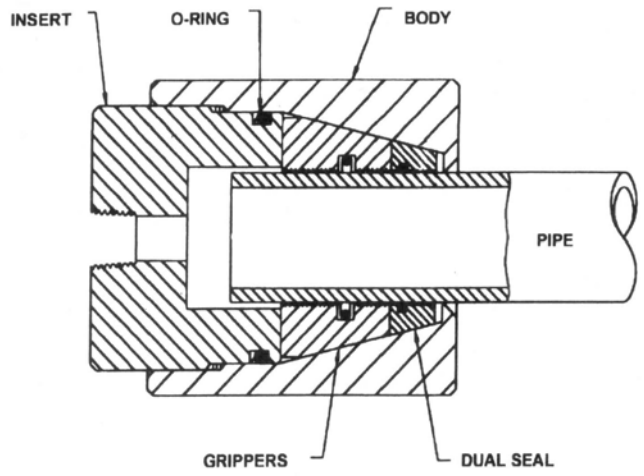


FIGURE 1. Typical OD hydro test plug (for information only).

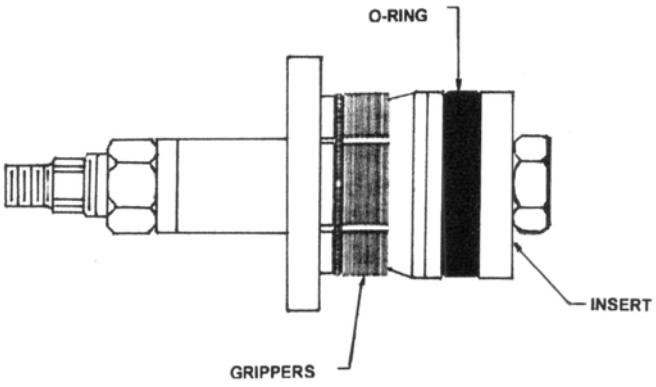


FIGURE 2. Typical ID hydro test plug (for information only).

3.4.2 Porting. The hydrostatic test plugs shall have a central port that connects the pressurized region of the pipe being hydrostatically tested to an outside, external fitting to which the hydrostatic pressure source pump can be connected. This fitting shall have an appropriately sized male or female pipe thread, conforming to ASME B1.20.1 tapered National Pipe Thread (NPT) inch-series threads or as specified by the buyer. The hydrostatic plug shall be supplied with a pipe cap or plug which can be used to seal the port.

3.4.3 Design of pressurized components. All components subjected to the hydrostatic pressure shall be designed in accordance with the ASME Boiler and Pressure Vessel Code, for a maximum allowable working pressure at least 1.5 times as great as the pressure rating stamped on the fitting. All components subjected to hydrostatic pressure shall be designed in accordance with the ASME Boiler and Pressure Vessel Code Section B31.1 or alternate method of design as approved by NAVSEA. The design shall be for the maximum allowable working pressure of the components using the material allowable stresses as dictated by the Code. Designs for which the maximum allowable working pressure cannot be computed with satisfactory assurance, refer to ASME Boiler and Pressure Vessel Code Section B31.1 or the ASME Boiler and Pressure Vessel Code Section VIII (Proof tests to establish maximum allowable working pressure). All hydrostatic test plugs shall be proven through hydrostatic testing using a test fixture representative of the intended final use. The test medium shall be fresh water at room temperature. The test shall be conducted after the fitting has been installed in the test fixture bore using the manufacturer's recommended tightening procedure. The test pressure shall be maintained for a minimum of 10 minutes. The fitting shall not blow out of the fixture or noticeably shift in position, and shall not leak.

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3.4.4 Pressure rating. The pressure rating stamped or marked on the hydrostatic test plug shall be at most the lesser of either two thirds the design pressure (see 3.4.3) or two thirds the pressure required to produce the holding load for which the plug is designed (see 3.4.6). The manufacturer has the option of using a pressure rating less than this, based on the maximum pressure rating of the pipe material, size, and schedule for which the hydrostatic test plug is specifically designed.

3.4.5 Gripper assemblies. Hydrostatic test plugs shall be provided with segmented, serrated gripper assemblies that are spring loaded or otherwise configured to be activated when tightening the hydro plug fittings and to readily relax when pressure is removed and the fitting loosened. The gripping assemblies shall contact and grip the inside diameter surface of the pipe, tube, or fitting in Type I designs, and the outside diameter surface in Type II designs. In the installation of the hydrostatic test plug in the pipe, tube, or fitting ,the gripper assemblies shall be set to the proper contact force against the surface by the tightening of the plug nuts or other fittings used for setting the fixture.

3.4.6 Holding load. The gripper assembly design shall be such that when the hydrostatic test plug is properly set in accordance with manufacturer's instructions, the entire holding load preventing blowout of the fitting at the maximum pressure, as defined in 3.4.4, will be provided by the gripper assembly contact force alone. Any holding force provided by the seal shall be extra holding force and shall not be required for keeping the plug from discharging from the pipe or fitting at the maximum qualification test pressure.

3.4.7 Insertion depth. Hydro plug assemblies shall be provided with dimensional offsets, as required, to ensure that the entire gripper assembly length will be contacting the pipe or fitting when the hydrostatic test plug is installed in any standard design pipe or fitting end.

3.4.8 Surface marking. Gripper elements shall not cause surface marking in pipe surfaces deeper than 10 mils.

3.4.9 Safety gag. At the manufacturer's option, a safety gag may be provided with the hydrostatic test plug that links the plug to a clamp that is secured on the pipe. If this gag is required by the manufacturer in order to ensure that the plug will not eject upon hydrostatic testing, the plug shall be marked indicating that the gag is required.

3.4.10 External pipe support clamps. When requested (see 6.3), the manufacturer shall supply clamped external pipe support rings to be used with Type I hydrostatic test plugs. These clamps shall be sized for the external diameter of the pipe, and shall be dimensioned so that the clamp abutting faces are flush at an external pipe diameter at the negative of the diametrical tolerance as given in table II. The support clamps shall be designed to prevent the flaring of the pipe by the Type I hydrostatic plug gripper element.

4. PRODUCT CONFORMANCE PROVISIONS.

4.1 Product conformance. The products provided shall meet the salient characteristics of this Commercial Item Description, conform to the producer's own drawings, specifications, standards, and quality assurance practices, and be the same product offered for sale in the commercial marketplace. The government reserves the right to require proof of such conformance.

5. PACKAGING. Preservation, packing, and marking shall be as specified in the contract or order.

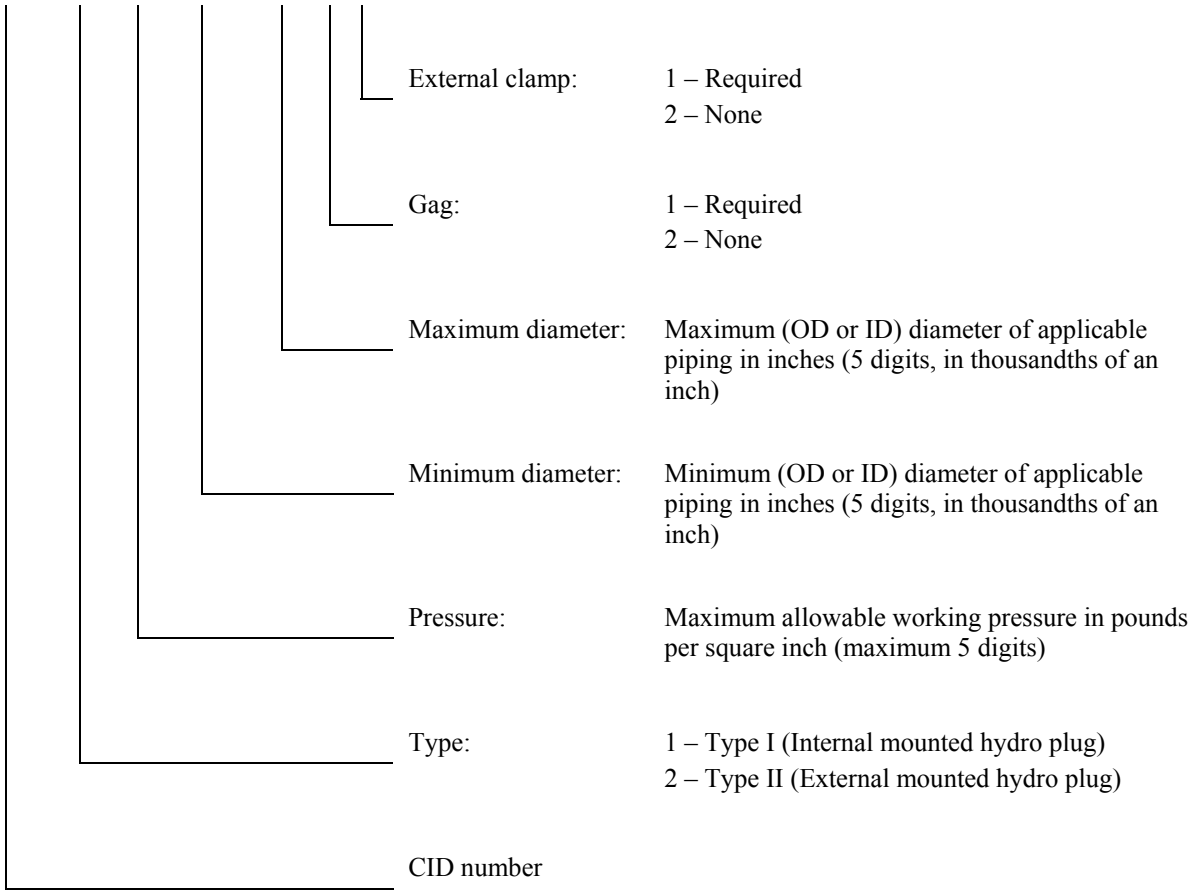
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6. NOTES.

6.1 Part or Identification Number (PIN). The following PIN procedure is for government purposes and does not constitute a requirement for the contractor.

This example describes a part numbering system for CID A-A-59822.

AA59822-1-03000-02440-02540-1-2



For example, AA59822-1-03000-02440-02540-1-2 represents an internal mounted hydro plug, 3000 pounds per square inch working pressure, a diameter of 2.440 inches to 2.540 inches, with a gag required, and no external clamp.

6.2 Source of documents.

6.2.1 ASME International. ASME International documents are available from ASME International, 22 Law Drive, PO Box 2900, Fairfield, NJ 07007-2900 or online at www.asme.org.

6.2.2 ASTM International. ASTM International documents are available from ASTM International, 100 Barr Harbor Dr., PO Box C700, West Conshohocken, PA 19428-2959 or online at www.astm.org.

6.2.3 Military specifications. Military specifications are available at http://assist.daps.dla.mil/quicksearch/ or http://assist.daps.dla.mil or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.

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6.3 Ordering data. The contract or order should specify the following:

a. CID document number, revision, and title.
b. Whether a Type I or Type II plug is required.
c. Product conformance provisions including certification documentation that all hydrostatic test plugs meet the pressure requirements.
d. Pipe diameter and schedule or class for standard pipe sizes.
e. Specific pipe outside diameter, inside diameter and/or wall thickness, for non-standard sizes.
f. Required minimum pressure rating.
g. Quantity required.
h. Whether an external pipe support ring is required with Type I plugs, and pipe diameter, if required (see 3.4.10).
i. Packaging, preservation, and marking requirements.

6.4 Potential sources. Based on information available at the time this document was developed, the following companies manufacture products that are believed to comply with the salient characteristics of this CID:

Expansion Seals Technology Group, Inc.
334 Godshall Drive
Harleysville, PA 19438-2008

Hy-Tech Machine, Inc.
Mashuda Industrial Park
25 Leonberg Road
Cranberry Twp, PA 16066

Sealfast Products
3501B S. 26th St.
Manitowoc, WI 54220

6.5 Key words.

Hydro

Hydrostatic test

Pipe plug

Pipe stopper

Test plug

MILITARY INTERESTS

Custodians:
Navy - SH
Air Force - 99

Review Activities:
Navy - AS, NP
Air Force - 03, 71
DLA - CC
GSA - FAS

CIVIL AGENCY COORDINATING ACTIVITY:
GSA - FAS

Preparing Activity:
Navy - SH
(Project 4730-2007-017)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at http://assist.daps.dla.mil.

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